INCH-POUND

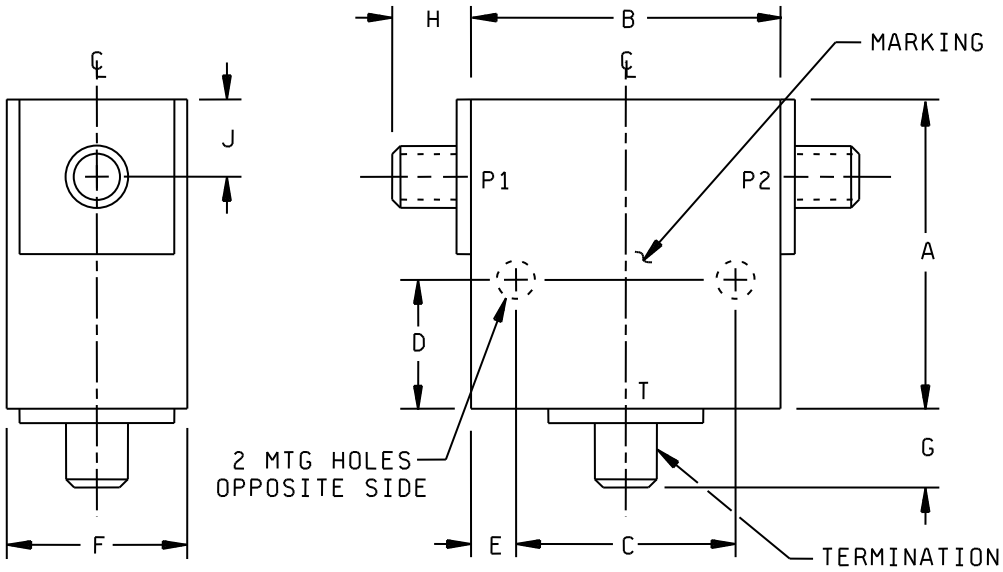
MIL-DTL-28791/1D
w/AMENDMENT 1
5 October 2020
SUPERSEDING
MIL-DTL-28791/1D
28 March 2016

DETAIL SPECIFICATION SHEET

ISOLATORS, RADIO FREQUENCY, COAXIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

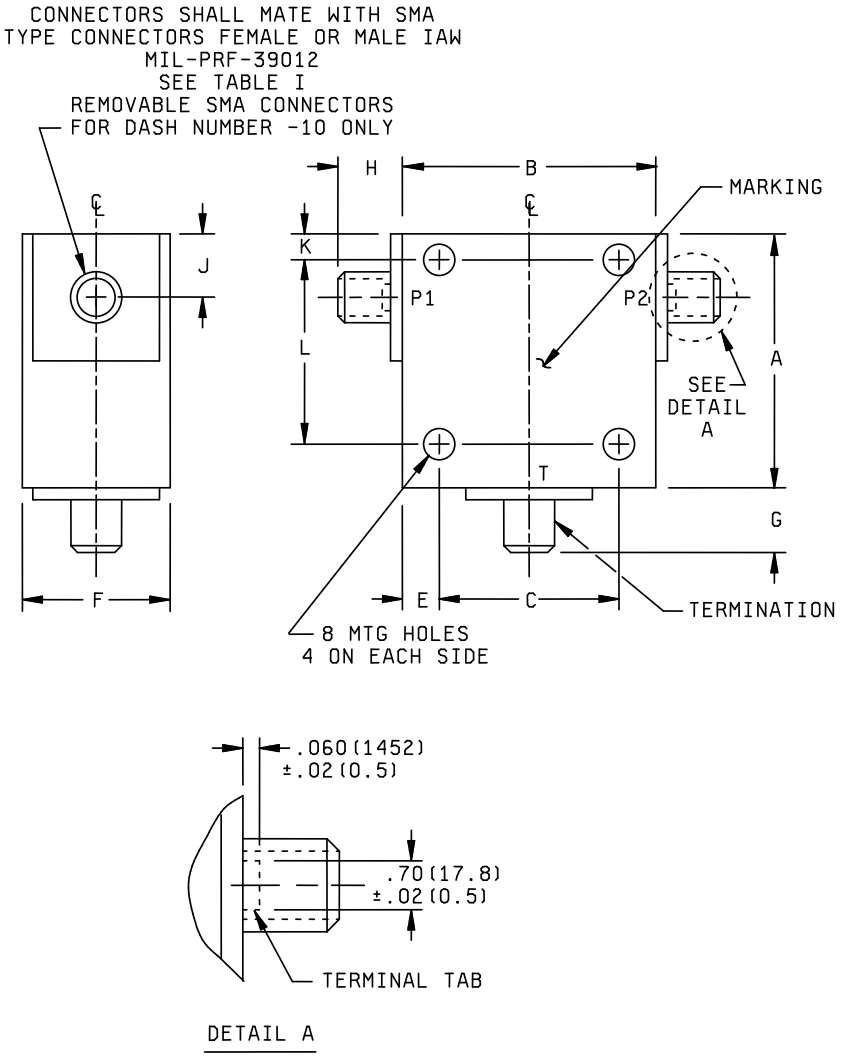
The requirements for acquiring the product listed herein shall consist of this specification sheet and MIL-DTL-28791.



NOTES:
1. See table I for dimensions.
2. All un-dimensioned pictorial representations are for reference purposes only.

FIGURE 1. Dimensions and configurations for dash numbers 001, 002, 003, 004, 005, 016, 017, 020, 022, 023, 024, 025, 030, 031, and 035.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.

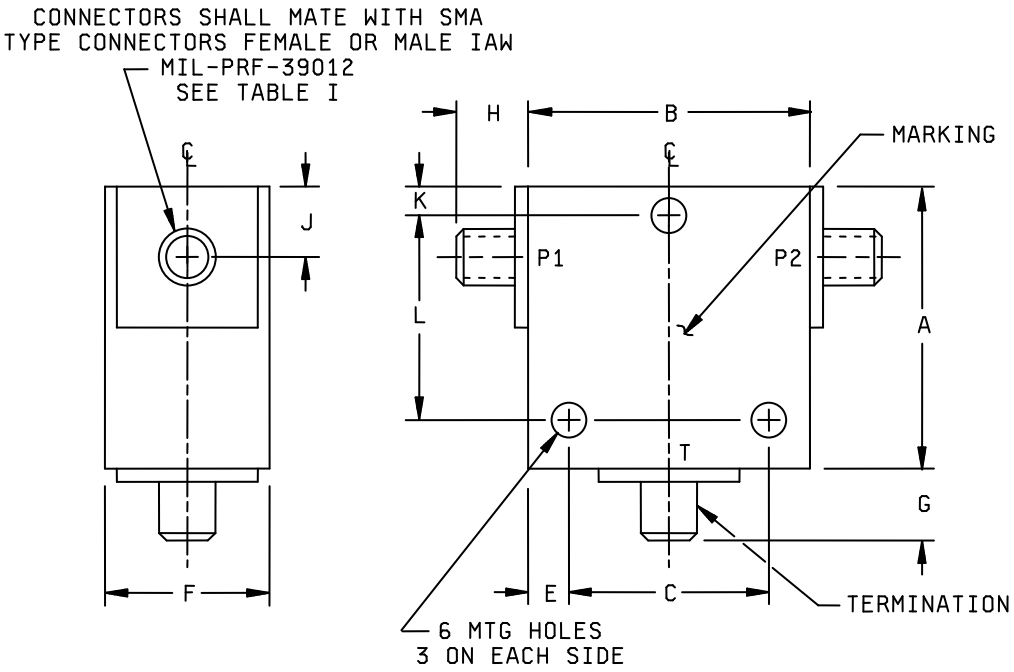


STRIPLINE TERMINAL TABS PROTRUSION FROM CASE WHEN INPUT AND OUTPUT PORT CONNECTORS ARE REMOVED. SHIM CONNECTOR MOUNTINGS AS REQUIRED.

NOTES:
1. See table I for dimensions.
2. All un-dimensioned pictorial representations are for reference purposes only.
3. Metric equivalents are in parentheses.
4. Dimensions are in inches.

FIGURE 2. Dimensions and configurations for dash numbers 006, 010, and 014.

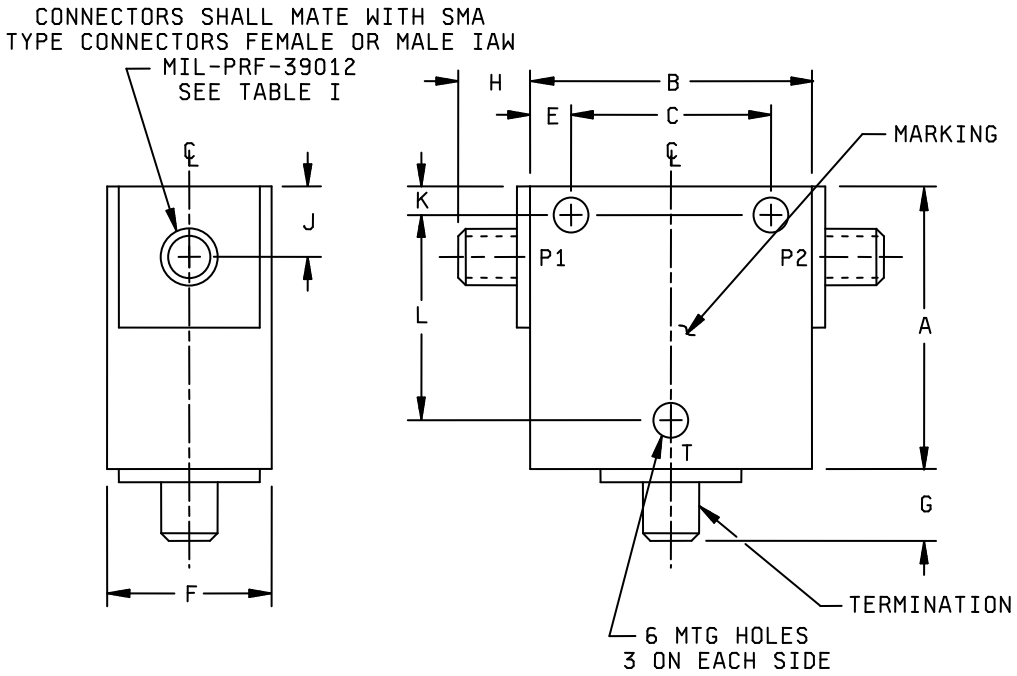
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.



NOTES:
1. See table I for dimensions.
2. All un-dimensioned pictorial representations are for reference purposes only.

FIGURE 3. Dimensions and configurations for dash numbers 007, 009, 011, 012, and 027.

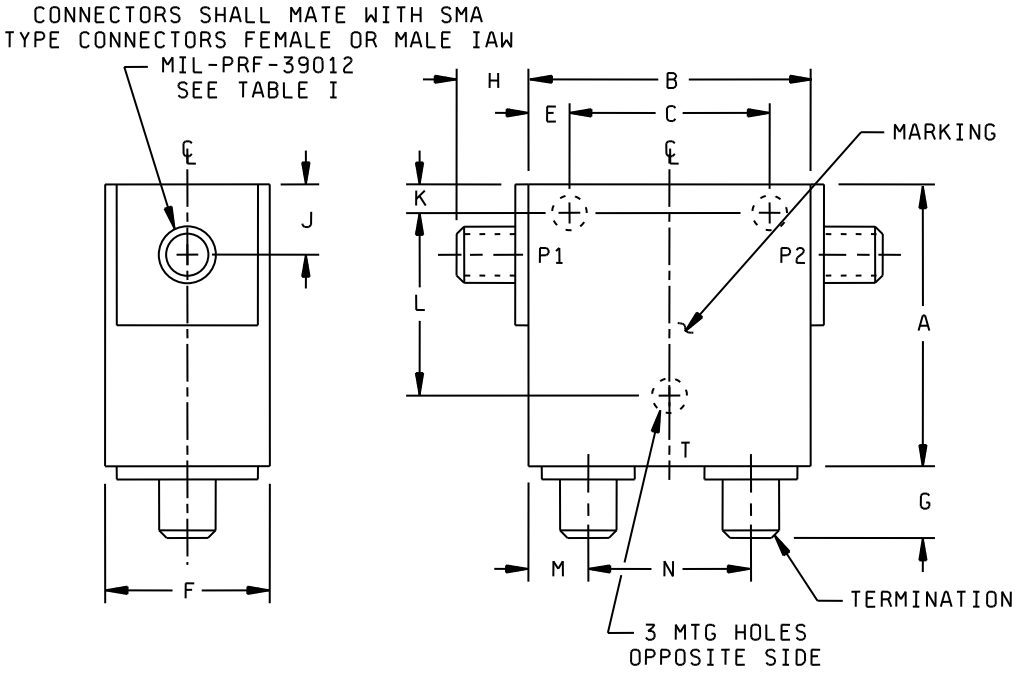
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.



NOTES:
1. See table I for dimensions.
2. All un-dimensioned pictorial representations are for reference purposes only.

FIGURE 4. Dimensions and configurations for dash numbers 008, 013, 015, 018, 019, 021, 028, and 033.

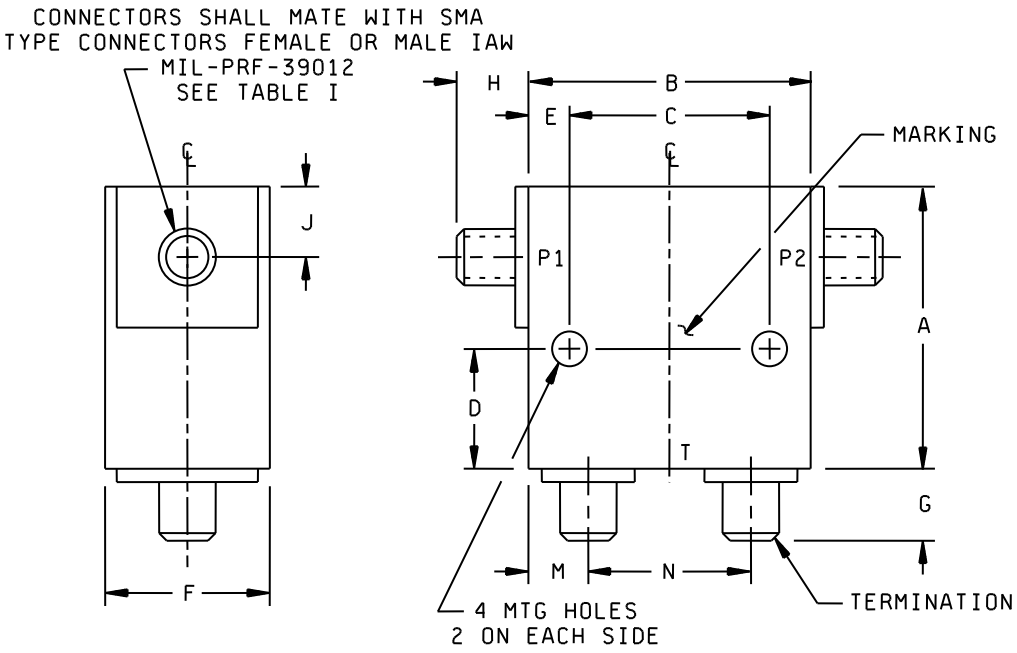
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.



NOTES:
1. See table I for dimensions.
2. All un-dimensioned pictorial representations are for reference purposes only.

FIGURE 5. Dimensions and configurations for dash numbers 026 and 032.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.



NOTES:
1. See table I for dimensions.
2. All un-dimensioned pictorial representations are for reference purposes only.

FIGURE 6. Dimensions and configurations for dash numbers 029 and 034.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensional and thread requirements.

Dash no.	Dimensions 1/ 2/ 3/													
	A	B	C		D		E		F	G	H		J	
	Max	Max	Min	Max	Min	Max	Min	Max	Max	Max	Min	Max	Min	Max
-001	.66 (16.8)	.53 (13.5)	.345 (8.76)	.355 (9.02)	.407 (10.34)	.417 (10.59)	.070 (1.78)	.080 (2.03)	.53 (13.5)	.135 (3.43)	.220 (5.59)	.415 (10.54)	.245 (6.22)	.255 (6.48)
-002	.66 (16.8)	.53 (13.5)	.345 (8.76)	.355 (9.02)	.407 (10.34)	.417 (10.59)	.070 (1.78)	.080 (2.03)	.53 (13.5)	.135 (3.43)	"	"	.245 (6.22)	.255 (6.48)
-003	1.055 (26.80)	.955 (24.26)	.645 (16.38)	.655 (16.64)	.395 (10.03)	.405 (10.29)	.145 (3.68)	.155 (3.94)	.65 (16.5)	.27 (6.9)	"	"	.245 (6.22)	.255 (6.48)
-004	1.055 (26.80)	.955 (24.26)	.645 (16.38)	.655 (16.64)	.395 (10.03)	.405 (10.29)	.145 (3.68)	.155 (3.94)	.65 (16.5)	.27 (6.9)	"	"	.245 (6.22)	.255 (6.48)
-005	1.20 (30.5)	1.20 (30.5)	.945 (24.00)	.955 (24.26)	.395 (10.03)	.405 (10.29)	.125 (3.18)	.135 (3.43)	.75 (19.0)	.17 (4.3)	"	"	.255 (6.48)	.265 (6.73)
-006	1.25 (31.8)	1.25 (31.8)	.645 (16.38)	.655 (16.64)			.295 (7.49)	.305 (7.75)	.75 (19.0)	.17 (4.3)	"	"	.295 (7.49)	.305 (7.75)
-007	1.50 (38.1)	1.50 (38.1)	.995 (25.27)	1.005 (25.53)			.245 (6.22)	.255 (6.48)	.63 (16.0)	.27 (6.9)	"	"	.265 (6.73)	.275 (6.98)
-008	2.75 (69.8)	2.75 (69.8)	1.995 (50.67)	2.005 (50.93)			.495 (12.57)	.505 (12.83)	.89 (22.6)	.69 (17.5)	"	"	.399 (10.13)	.349 (8.86)
-009	1.25 (31.8)	1.25 (31.8)	.995 (25.27)	1.005 (25.53)			.115 (2.92)	.125 (3.18)	.63 (16.0)	.27 (6.9)	"	"	.245 (6.22)	.255 (6.48)
-010	1.500 (38.10)	1.49 (37.8)	.995 (25.27)	1.005 (25.53)			.245 (6.22)	.255 (6.48)	.65 (16.5)	.17 (4.3)	"	"	.275 (6.98)	.285 (7.24)
-011	1.25 (31.8)	1.25 (31.8)	.995 (25.27)	1.005 (25.53)			.115 (2.92)	.125 (3.18)	.63 (16.0)	.27 (6.9)	"	"	.245 (6.22)	.255 (6.48)
-012	2.00 (50.8)	2.00 (50.8)	1.713 (43.51)	1.723 (43.76)			.135 (3.43)	.145 (3.68)	1.00 (25.4)	.27 (6.9)	"	"	.495 (12.57)	.505 (12.83)
-013	1.70 (43.2)	1.63 (41.4)	1.115 (28.32)	1.125 (28.58)			.245 (6.22)	.255 (6.48)	.75 (19.0)	.27 (6.9)	"	"	.349 (8.86)	.355 (9.02)
-014	1.72 (43.7)	1.65 (41.9)	1.045 (26.54)	1.055 (26.80)			.295 (7.49)	.305 (7.75)	.75 (19.0)	.17 (4.3)	"	"	.335 (8.51)	.345 (8.76)
-015	2.08 (52.8)	1.77 (45.0)	1.095 (27.81)	1.105 (28.07)			.325 (8.26)	.335 (8.51)	.99 (25.1)	.17 (4.3)	"	"	.275 (6.98)	.285 (7.24)
-016	1.625 (41.28)	1.625 (41.28)	1.320 (33.53)	1.330 (33.78)	.807 (20.50)	.817 (20.75)	.120 (3.05)	.130 (3.30)	.75 (19.0)	.27 (6.9)	"	"	.370 (9.40)	.380 (9.65)
-017	1.625 (41.28)	1.625 (41.28)	1.320 (33.53)	1.330 (33.78)	.807 (20.50)	.817 (20.75)	.120 (3.05)	.130 (3.30)	.75 (19.0)	.27 (6.9)	"	"	.370 (9.40)	.380 (9.65)

See footnotes at end of table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensional and thread requirements - Continued.

Dash no.	Dimensions - Continued. 1/ 2/ 3/						
	K		L		MTG threads	Connectors	Figure no.
	Min	Max	Min	Max			
-001					.086-56 UNC-2B x.13DP	2 females	1
-002					.086-56 UNC-2B x .13DP	2 females	1
-003					.112-40 UNC-2B x .15DP	2 females	1
-004					.112-40 UNC-2B x .15DP	2 females	1
-005					.086-56 UNC-2B x .13DP	2 females	1
-006	.145 (3.68)	.155 (3.94)	.995 (25.27)	1.005 (25.53)	.086-56 UNC-2B x .30DP	2 females	2
-007	.145 (3.68)	.155 (3.94)	1.182 (30.02)	1.192 (30.28)	.112-40 UNC-2B x .13DP	2 females	3
-008	.399 (10.13)	.349 (8.86)	1.995 (50.67)	2.005 (50.93)	.112-40 UNC-2B x .19DP	2 females	4
-009	.115 (2.92)	.125 (3.18)	.995 (25.27)	1.005 (25.53)	.086-56 UNC-2B x .13DP	1 female and 1 male	3
-010	.115 (2.92)	.125 (3.18)	.995 (25.27)	1.005 (25.53)	.086-56 UNC-2B x .13DP	2 females	2
-011	.115 (2.92)	.125 (3.18)	.995 (25.27)	1.005 (25.53)	.086-56 UNC-2B x .13DP	1 female and 1 male	3
-012	.135 (3.43)	.145 (3.68)	1.713 (43.51)	1.723 (43.76)	.112-40 UNC-2B x .15DP	2 females	3
-013	1.075 (27.30)	1.085 (27.56)	.349 (8.86)	.355 (9.02)	.112-40 UNC-2B x .15DP	2 females	4
-014	.335 (8.51)	.345 (8.76)	1.0495 (26.657)	1.0505 (26.683)	.112-40 UNC-2B x .15DP	2 females	2
-015	.345 (8.76)	.355 (9.02)	1.395 (35.43)	1.405 (35.69)	.112-40 UNC-2B x .15DP	2 females	4
-016					.112-40 UNC-2B x .15DP	2 females	1
-017					.112-40 UNC-2B x .15DP	2 females	1

See footnotes at end of table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensional and thread requirements - Continued.

Dash no.	Dimensions 1/ 2/ 3/													
	A	B	C		D		E		F	G	H		J	
	Max	Max	Min	Max	Min	Max	Min	Max	Max	Max	Min	Max	Min	Max
-018	1.00 (25.4)	1.00 (25.4)	.795 (20.19)	.805 (20.45)			.095 (2.41)	.105 (2.67)	.50 (12.7)	.17 (4.3)	.220 (5.59)	.415 (10.54)	.245 (6.22)	.255 (6.48)
-019	1.06 (26.9)	1.00 (25.4)	.745 (18.92)	.755 (19.18)			.115 (2.92)	.125 (3.18)	.75 (19.0)	.27 (6.9)	"	"	.245 (6.22)	.255 (6.48)
-020	1.03 (26.2)	1.00 (25.4)	.595 (15.11)	.605 (15.37)	.395 (10.03)	.405 (10.29)	.195 (4.95)	.205 (5.21)	.63 (16.0)	.17 (4.3)	"	"	.275 (6.98)	.285 (7.24)
-021	1.30 (33.0)	1.15 (29.2)	.695 (17.65)	.705 (17.91)			.225 (5.72)	.235 (5.97)	.63 (16.0)	.17 (4.3)	"	"	.275 (6.98)	.285 (7.24)
-022	1.00 (25.4)	1.00 (25.4)	.745 (18.92)	.755 (19.18)	.182 (4.62)	.192 (4.88)	.120 (3.05)	.130 (3.30)	.50 (12.7)	.27 (6.9)	"	"	.245 (6.22)	.255 (6.48)
-023	1.00 (25.4)	1.00 (25.4)	.745 (18.92)	.755 (19.18)	.182 (4.62)	.192 (4.88)	.120 (3.05)	.130 (3.30)	.50 (12.7)	.27 (6.9)	"	"	.245 (6.22)	.255 (6.48)
-024	1.12 (28.4)	.91 (23.1)	.820 (20.83)	.830 (21.08)	.090 (2.29)	.100 (2.54)	.080 (2.03)	.090 (2.29)	.90 (22.9)	.27 (6.9)	"	"	.375 (9.52)	.385 (9.78)
-025	.85 (21.6)	.75 (19.0)	.595 (15.11)	.605 (15.37)	.255 (6.48)	.265 (6.73)	.075 (1.90)	.085 (2.16)	.60 (15.2)	.17 (4.3)	"	"	.255 (6.48)	.265 (6.73)
-026	.85 (21.6)	1.60 (40.6)	1.395 (35.43)	1.405 (35.69)			.095 (2.41)	.105 (2.67)	.60 (15.2)	.17 (4.3)	"	"	.245 (6.22)	.255 (6.48)
-027	1.06 (26.9)	.87 (22.1)	.620 (15.75)	.630 (16.00)			.115 (2.92)	.125 (3.18)	.81 (20.6)	.17 (4.3)	"	"	.245 (6.22)	.255 (6.48)
-028	1.08 (27.4)	.94 (23.9)	.620 (15.75)	.630 (16.00)			.155 (3.94)	.165 (4.19)	.56 (14.2)	.17 (4.3)	"	"	.275 (6.98)	.285 (7.24)
-029	1.00 (25.4)	1.80 (45.7)	1.595 (40.51)	1.605 (40.77)	.375 (9.52)	.385 (9.78)	.095 (2.41)	.105 (2.67)	.63 (16.0)	.17 (4.3)	"	"	.265 (6.73)	.275 (6.98)
-030	.77 (19.6)	.63 (16.0)	.495 (12.57)	.505 (12.83)	.515 (13.08)	.525 (13.34)	.058 (1.47)	.068 (1.73)	.50 (12.7)	.25 (6.4)	"	"	.245 (6.22)	.255 (6.48)
-031	.84 (21.3)	.63 (16.0)	.465 (11.81)	.475 (12.06)	.325 (8.26)	.335 (8.51)	.075 (1.90)	.085 (2.16)	.62 (15.7)	.17 (4.3)	"	"	.285 (7.24)	.295 (7.49)
-032	.68 (17.3)	1.02 (25.9)	.855 (21.72)	.865 (21.97)			.075 (1.90)	.085 (2.16)	.56 (14.2)	.45 (11.4)	"	"	.265 (6.73)	.275 (6.98)
-033	.83 (21.1)	.67 (17.0)	.355 (9.02)	.365 (9.27)			.155 (3.94)	.165 (4.19)	.62 (15.7)	.35 (8.9)	"	"	.275 (6.98)	.285 (7.24)
-034	.78 (19.8)	1.20 (30.5)	.955 (24.26)	.965 (24.51)	.525 (13.34)	.535 (13.59)	.115 (2.92)	.125 (3.18)	.63 (16.0)	.15 (3.8)	"	"	.245 (6.22)	.255 (6.48)
-035	.77 (19.6)	.63 (16.0)	.495 (12.57)	.505 (12.83)	.515 (13.08)	.525 (13.34)	.058 (1.47)	.068 (1.73)	.80 (20.3)	.25 (6.4)	"	"	.245 (6.22)	.255 (6.48)

See footnotes at end of table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensional and thread requirements - Continued.

Dash no.	Dimensions - Continued. 1/ 2/ 3/								
	K		L		M	N	MTG threads	Connectors	Fig. no.
	Min	Max	Min	Max	Max	Max			
-018	.225 (5.72)	.235 (5.97)	.645 (16.38)	.655 (16.64)			.086-56 UNC-2B x .13DP	2 females	4
-019	.225 (5.72)	.235 (5.97)	.695 (17.65)	.705 (17.91)			.086-56 UNC-2B x .13DP	2 females	4
-020							.112-40 UNC-2B x .15DP	2 females	1
-021	.175 (4.44)	.185 (4.70)	.945 (24.00)	.955 (24.26)			.112-40 UNC-2B x .15DP	2 males	4
-022							.125 DIA through	2 females	1
-023							.125 DIA through	2 females	1
-024							.086-56 UNC-2B x .13DP	2 females	1
-025							.086-56 UNC-2B x .16DP	2 females	1
-026	.245 (6.22)	.255 (6.48)	.495 (12.57)	.505 (12.83)	.40 (10.2)	.80 (20.3)	.086-56 UNC-2B x .16DP	2 females	5
-027	.145 (3.68)	.155 (3.94)	.732 (18.59)	.742 (18.85)			.112-40 UNC-2B x .18DP	2 females	3
-028	.245 (6.22)	.255 (6.48)	.595 (15.11)	.605 (15.37)			.086-56 UNC-2B x .16DP	2 males	4
-029					.45 (11.4)	.45 (11.4)	.112-40 UNC-2B x .15DP	2 females	6
-030							.086-56 UNC-2B x .13DP	2 females	1
-031							.086-56 UNC-2B x .13DP	2 females	1
-032	.265 (6.73)	.275 (6.98)	.305 (7.75)	.315 (8.00)	.25 (6.4)	.51 (13.0)	.086-56 UNC-2B x .13DP	2 females	5
-033	.225 (5.72)	.235 (5.97)	.485 (12.32)	.495 (12.57)			.086-56 UNC-2B x .16DP	2 males	4
-034							.086-56 UNC-2B x .16DP	2 females	6
-035							.086-56 UNC-2B x .13DP	2 females	1

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only.
3/ Metric equivalents are in parentheses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.

Part or Identifying Number (PIN): M28791/1 - (dash number from table II).

ENGINEERING DATA:

Frequency range: See table II.

Voltage rating (sea level): 335 V.

Nominal impedance: 50 ohms.

Operating temperature range: See table II.

Power rating: See table II.

Input port: P_1.

Output port: P_2.

RF connectors: Shall be stainless steel.

REQUIREMENTS:

Dimensions and configurations: See figure 1 through figure 6, and table I.

Dielectric withstanding voltage: 1,000 V rms minimum.

Isolation: See table II.

Insertion loss: See table II.

VSWR: See table II.

Marking: See figure 1 through figure 6.

Salt atmosphere (where applicable): MIL-STD-202-101, test condition B. This test shall be performed under first article inspection after the vibration test. Insertion loss and VSWR shall be measured after the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical and additional environmental requirements.

Dash number	Frequency range	Isolation	Insertion loss	VSWR	Operating temperature range	Input power rating average	Salt atmosphere 1/
	GHz	dB min	dB max	max	°C	Watt min	
-001	9.50-10.0	20	0.3	1.25:1	-55°C to +95°C	10	N/A
-002	12.5-13.5	20	0.3	1.25:1	-55°C to +95°C	10	N/A
-003	7.0-11.0	20	0.5	1.30:1	-55°C to +95°C	10	N/A
-004	11.0-18.0	20	0.5	1.30:1	-55°C to +95°C	10	N/A
-005	0.95-1.225	20	0.5	1.25:1	-54°C to +85°C	2	N/A
-006	0.95-1.225	22	0.4	1.25:1	0°C to +71°C	2	Condition B
-007	0.96-1.215	18	0.5	1.25:1		75	N/A
-008	1-2	18	0.5	1.30:1	0°C to +71°C	2	Condition B
-009	1.255-1.41	20	0.4	1.25:1	-28°C to +65°C	5	Condition B
-010	1.26-1.666	20	0.5	1.25:1	-40°C to +65°C	0.03	Condition B
-011	1.495-1.65	20	0.3	1.25:1	-28°C to +65°C	5	Condition B
-012	1.53-1.87	20	0.5	1.25:1	-20°C to +70°C	2	N/A
-013	1.8-4.2	20	0.8	1.50:1	-40°C to +75°C	2	Condition B
-014	2-4	20	0.4	1.25:1	-55°C to +125°C	1	Condition B
-015	2-4	20	0.4	1.25:1	-54°C to +71°C	2	Condition B
-016	2.16-4.16	18	0.5	1.35:1	-55°C to +85°C	2	Condition B
-017	2.2-4.2	18	0.5	1.35:1	-55°C to +85°C	2	Condition B
-018	2.7-2.9	20	0.4	1.25:1	-41°C to +53°C	2	Condition B
-019	3.6-8.4	20	0.5	1.50:1	-40°C to +75°C	2	Condition B
-020	4.0-8.0	20	0.5	1.30:1	-55°C to +125°C	1	Condition B
-021	4.0-8.0	20	0.4	1.25:1	-54°C to +70°C	2	Condition B
-022	4.16-8.16	18	0.5	1.35:1	-55°C to +85°C	2	Condition B
-023	4.2-8.2	18	0.5	1.35:1	-55°C to +85°C	2	Condition B
-024	4.35-4.85	25	0.5	1.18:1	-40°C to +71°C	2	N/A
-025	7-11	20	0.4	1.22:1	-54°C to +105°C	10	N/A
-026	7.55-12.85	25	1.2	1.35:1	-40°C to +75°C	2	Condition B
-027	8-12	20	0.4	1.25:1	-54°C to +70°C	1	N/A
-028	8-12	20	0.4	1.25:1	-55°C to +95°C	2	Condition B
-029	8-12	40	0.8	1.25:1	0°C to +65°C	2	Condition B
-030	8.16-12.56	18	0.5	1.35:1	-55°C to +85°C	2	Condition B
-031	8-16	17	0.7	1.35:1	-45°C to +85°C	2	Condition B
-032	12-18.5	30	1.2	1.35:1	-40°C to +75°C	2	Condition B
-033	12-18	18	0.6	1.30:1	-54°C to +71°C	2	Condition B
-034	12.4-18	40	0.8	1.25:1	0°C to +65°C	2	Condition B
-035	12.56-18.16	17	1.0	1.45:1	-55°C to +85°C	2	Condition B

1/ This test shall be performed under first article inspection and be conducted after the vibration test. After exposure, isolators shall be examined for evidence of corrosion. Any corrosion shall be considered a failure. After the salt atmosphere test, the VSWR shall be measured and shall not be greater than specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.

Referenced documents. In addition to MIL-DTL-28791, this specification sheet references the following documents:
MIL-PRF-39012
MIL-STD-202-101

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2020-018)

Review activities:
Navy - AS, MC, OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:48Z
Check the source to verify that this is the current version before use.